

9(a)	any seven from: 1 light absorbed by, chlorophyll / pigments (in PI and PII) or photoactivation of chlorophyll (in PI and PII) ; 2 electrons, emitted (to higher energy levels) / excited, in both photosystems ; 3 electrons pass along, electron transport chain / ETC / electron carriers ; 4 detail ; e.g. electrons releasing energy to pump H⁺ into thylakoid, space / lumen <i>ref. to</i> proton gradient set up protons diffusing through ATP synthase 5 ATP synthesised from ADP and P_i / <i>ref. to</i> chemiosmosis ; 6 (PII contains) oxygen-evolving complex ; A water-splitting enzyme 7 <i>ref. to</i> photolysis (of water) ; 8 oxygen and hydrogen ions and electrons produced ; A $\text{H}_2\text{O} \rightarrow 2\text{e}^- + 2\text{H}^+ + \frac{1}{2}\text{O}_2$ 9 electrons emitted from PI replaced by electrons from PII or electrons emitted from PII replaced by electrons from photolysis (of water) ; 10 electrons (from PI) combine with H⁺ to form reduced NADP or electrons (from PI) combine with NADP to form reduced NADP or electrons (from PI) combine with H⁺ to form hydrogen that reduces NADP ; R H⁺ reduce NADP	7
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Question	Answer	Marks
9(b)	1 increase in light intensity increases rate of photosynthesis until it levels off (at higher light intensities) ; 2 (increase means) more, light dependent reaction / cyclic photophosphorylation / non-cyclic photophosphorylation / photoactivation ; 3 (as rate levels off) carbon dioxide concentration / temperature, becomes limiting factor or light <u>intensity</u> is no longer a limiting factor ;	3

Question	Answer	Marks
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